

3 Human Immunodeficiency Virus (HIV) infects human cells, causing symptoms that may result in death.

(a) Explain why the virus is an immunodeficiency virus.

(4)

[illegible]

(b) A number of antiretroviral drugs are given to patients with HIV.

These include reverse transcriptase inhibitors and integrase inhibitors.

(i) Explain how these two drugs reduce the development of HIV.

(2)

- (ii) These two drugs are often given in combination with other antiretroviral drugs.

Suggest why patients are given combinations of drugs.

(2)

- (iii) If a patient stops taking the antiretroviral drugs, the number of virus particles increases again.

Suggest why this increase occurs.

(2)

(Total for Question 3 = 10 marks)